

习题一

6. 解: 1) $L_0(x) = \frac{1}{2}(x-1)$ $L_1(x) = \frac{1}{2}(x+1)$ $y_0 = -1$ $y_1 = 1$

$$\therefore P_1(x) = L_0(x)y_0 + L_1(x)y_1 = x$$

2) $y_0 = -1$, $y_1 = 0$, $y_2 = 1$

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{1}{2}x(x-1)$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = 1-x^2$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{1}{2}x(x+1)$$

$$P_2(x) = \frac{1}{2}x(x-1) \cdot (-1) + (1-x^2) \cdot 0 + \frac{1}{2}x(x+1) \cdot 1$$

$$= x$$

3) $y_3 = 8$

$$L_0(x) = \frac{x(x-1)(x-2)}{(-1) \cdot (-2) \cdot (-3)} = -\frac{1}{6}x(x-1)(x-2)$$

$$L_1(x) = \frac{(x+1) \cdot x \cdot (x-2)}{2 \cdot 1 \cdot (-1)} = -\frac{1}{2}x(x+1)(x-2)$$

$$L_2(x) = \frac{(x+1) \cdot x \cdot (x-1)}{1 \cdot 2 \cdot 1} = \frac{1}{6}x(x+1)(x-1)$$

$$P_3(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x) = -\frac{1}{6}x^3 + \frac{1}{2}x^2 + x$$

$$= -\frac{1}{6}x^3 + \frac{1}{2}x^2 + x + (-\frac{1}{2}x^3 + \frac{1}{2}x^2 - \frac{1}{2}x) + 8x(\frac{1}{6}x^3 - \frac{1}{6}x)$$

$$= x^3$$

11. 取 $(x_0, y_0), (x_1, y_1), (x_2, y_2)$ 为样本点

$$l_0(x) = \frac{(x-0.47)(x-0.48)}{0.01 \times 0.02}$$

$$l_1(x) = \frac{(x-0.46)(x-0.47)}{0.01 \times (-0.01)}$$

$$l_2(x) = \frac{(x-0.46)(x-0.47)}{0.02 \times 0.01}$$

$$p(x) = l_0(x)y_0 + l_1(x)y_1 + l_2(x)y_2$$

$$\text{令 } x = 0.472, \text{ 得 } p(x) = 0.495573352$$

12) 取 $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ 为样本点

$$l_1(x) = \frac{(x-0.48)(x-0.49)}{0.01 \times 0.02}$$

$$l_2(x) = \frac{(x-0.47)(x-0.49)}{0.01 \times (-0.01)}$$

$$l_3(x) = \frac{(x-0.47)(x-0.48)}{0.02 \times 0.01}$$

$$p(x) = l_1(x)y_1 + l_2(x)y_2 + l_3(x)y_3$$

$$\text{令 } p(x) = 0.5, \text{ 解得 } x = 0.495553008$$

12. 解: 由拉格朗日插值余项公式得

$$R_n(x) = f(x) - p_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \prod_{k=0}^n (x-x_k)$$

$$R_3(x) = \frac{f^{(4)}(\xi)}{4!} \prod_{k=0}^3 (x-x_k)$$

$$(1) f(x) = 4x^3 - 3x + 2 \quad f^{(4)}(x) = 0$$

$$\therefore R_3(x) = 0$$

$$(2) f(x) = x^4 - 2x^3 \quad f^{(4)}(x) = 4!$$

$$R_3(x) = \frac{4!}{4!} (x+1)(x-7)(x-3)(x-4)$$

13. 解: (1) 线性插值

因为 $x = 0.3367$ 在节点 x_0 和 x_1 之间, 先估计误差

$$R_1(x) = \frac{f''(\xi)}{2!} (x-x_0)(x-x_1) = \frac{\sin(\xi)}{2} (x-x_0)(x-x_1) \leq \max_{\xi} \frac{(x-x_0)(x-x_1)}{2} = \frac{0.01^2}{2} = \frac{1}{2} \times 10^{-4}$$

须保留到小数点后4位, 计算时多取两位

$$P_1(x) = \frac{1}{x_1 - x_0} [(x-x_0) \sin(x_1) + (x_1-x) \sin(x_0)]$$

$$= \frac{1}{0.01} [(0.3367-0.32) \sin(0.34) + (0.34-0.3367) \sin(0.32)]$$

$$\approx 0.330365$$

14.

$$12) R_2(x) = \frac{f^{(4)}(\xi)}{3!} (x-x_0)(x-x_1)(x-x_2) = \frac{-1015}{6} (x-x_0)(x-x_1)(x-x_2) \leq \frac{\max |x-x_0|(x-x_1)(x-x_2)|}{6}$$

$$\approx \frac{3 \times 0.01^3}{6} = \frac{1}{2} \times 10^{-6}$$

$$P_2(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} \sin(x_0) + \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} \sin(x_1) + \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} \sin(x_2)$$

$$P_2(0.3367) = \frac{10^{-5}}{0.02^2} [3.8445 \times \sin(10.32) + 3.8911 \times \sin(10.34) - 2.7555 \times \sin(10.36)]$$

$$\approx 0.330374$$

则线性插值误差约为 0.92×10^{-5} , 二次插值误差为 0.24×10^{-6}

16. 解: $f(1)=7$ $f(2)=21$ $f(3)=59$ $f(4)=133$ $f(5)=255$

$$f(1,2) = \frac{f(2)-f(1)}{2-1} = 14 \quad f(2,3) = \frac{f(3)-f(2)}{3-2} = 38$$

$$f(3,4) = \frac{f(4)-f(3)}{4-3} = 74 \quad f(4,5) = \frac{f(5)-f(4)}{5-4} = 122$$

$$f(1,2,3) = \frac{f(2,3)-f(1,2)}{3-1} = \frac{38-14}{2} = 12$$

$$f(2,3,4) = \frac{74-38}{2} = 18$$

$$f(3,4,5) = \frac{122-74}{2} = 24$$

$$f(1,2,3,4) = \frac{f(2,3,4)-f(1,2,3)}{4-1} = \frac{18-12}{3} = 2$$

$$f(2,3,4,5) = \frac{24-18}{2} = 3$$

$$f(1,2,3,4,5) = 0$$

17. 解:

x	$f(x_0, \cancel{x})$	$f(x_0, x_1, \cancel{x})$	$f(x_0, x_1, x_2, \cancel{x})$	$f(x_0, x_1, x_2, x_3)$	$f(x_0, x_1, x_2, x_3, x_4)$	
1	-3					
2	0	3				
3	15	15	6			
4	48	33	9	1		
5	105	57	12	-1	0	
6	192	87	15	1	0	1.6

$$p = -3 + 3(x-1) + 6(x-2)(x-1) + (x-3)(x-2)(x-1)$$

31. 解: 由定理5可得 $|f(x) - S_1(x)| \leq \frac{h^2}{8} \max_{0 \leq x \leq \pi} |f''(x)| = \frac{h^2}{8}$

则 $\frac{h^2}{8} \leq \frac{1}{2} \times 10^{-5}$, 即 $h \leq \sqrt{40} \times 10^{-3}$

36. 解: $\bar{x}_T = 0$, $\bar{y}_T = 18.1183$ $\bar{x}_T^2 = 375$ $\bar{x}_T^3 = 0$, $\bar{x}_T^4 = 2.765625$

$\sum x_i y_i = 8.452275$, $\sum x_i^2 y_i = 7.58526875$

(1) - 1次

$$\begin{cases} aN + b\bar{x}_T = \bar{y}_T \\ a\bar{x}_T + b\bar{x}_T^2 = \sum x_i y_i \end{cases} \Rightarrow \begin{cases} a = 2.01314 \\ b = 2.25165 \end{cases}$$

$\therefore y = 2.01314 + \cancel{2.24165}x + 2.25165x$

(2) - 2次

$$\begin{cases} \cancel{a_0} + aN + b\bar{x}_T + c\bar{x}_T^2 = \bar{y}_T \\ a\bar{x}_T + b\bar{x}_T^2 + c\bar{x}_T^3 = \sum x_i y_i \\ a\bar{x}_T^2 + b\bar{x}_T^3 + c\bar{x}_T^4 = \sum x_i^2 y_i \end{cases} \Rightarrow \begin{cases} a = 2.0001 \\ b = 2.25165 \\ c = 0.0313 \end{cases}$$

$\therefore y = 2.0001 + 2.25165x + 0.0313x^2$

37. 解: 令 $X = x^2$, 则 $y = a + bX$ 为线性拟合, 取 $m=2$, $N=5$, 有下列方程组:

$$\begin{cases} 5a + b \sum_{i=1}^5 x_i = 5a + b \sum_{i=1}^5 x_i^2 = \sum_{i=1}^5 y_i \\ a \sum_{i=1}^5 x_i + b \sum_{i=1}^5 x_i^2 = a \sum_{i=1}^5 x_i^2 + b \sum_{i=1}^5 x_i^4 = \sum_{i=1}^5 x_i y_i = \sum_{i=1}^5 x_i^2 y_i \end{cases}$$

列出下表有: $\sum x_i = 157$ $\sum y_i = 271.4$ $\sum x_i^2 = 5327$

$$\sum x_i^4 = 7217699 \quad \sum x_i^2 y_i = 369321.5$$

将所求系数代入方程组, 得

$$\begin{cases} a \approx \cancel{0.47528} 0.97258 \\ b = 0.05004 \end{cases}$$

$$\text{即 } y = \cancel{0.475} 0.97258 + 0.05004 x^2$$